## Embedded Real Time Systems

# Requirements Review

Mandatory Exercises 3, 4, 5 and 7

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Review snapshot: `assignment_2/Firas`, commit `c38a216`.

## Post-review closure: 11 October 2026

This summary covers local fixes on `assignment_2/Firas` after the reviewed commit. The original PDFs and LaTeX sources were preserved under `outputs/fix_evidence/pre_fix_documents/`.

| Review finding | Implemented correction and evidence |
|---|---|
| Ex3 LED/GPIO mismatch | Custom LED register-0 writes and semantic SDT/legacy switch lookup. Native build and physical UART pass. |
| Ex3 OCM requirement | Canonical linker script explicitly selected; each actual ARM ELF has heap `0x1000-0x2FFF`, stack `0x3000-0x67FF`; driver configuration section retained. |
| Ex3 counter | UART records every value 0–15. First-run arrival spacing is 0.992–1.008 seconds. User confirmed the full physical LED count during rerun. |
| Ex4 interfaces/fixture | Course `vectorType`/`VectorArray` and function signatures restored; literal transpose gives repeated row values 30/70/110/150. Host tests and native Ex4 run pass. |
| Ex5 missing measurements | Three real-board comparisons pass both equality and known matrix values. Software ticks: 264, 197, 192; hardware: 3190, 3116, 3122. |
| Ex7 reference discrepancy | Original kept unchanged; explicitly adapted course reference added with four additional signed/zero/extreme cases. Literal untouched-testbench acceptance is still unresolved. |
| Benchmark/build robustness | Application `-O2`; shared ANN dimensions, driver/address guards, padded transfers, full transaction timing and ten trials. Workspaces preserved by rename and board cable selected explicitly. |

The Ex345 native bitstream met timing (WNS +1.216 ns, WHS +0.046 ns, zero negative slack) and final DRC reported zero violations. Original DDR skew warnings remain visible in the build log rather than being suppressed.

### Remaining acceptance boundaries

The course specifies 2017.2; native success in 2026.1 does not establish course approval. The original ANN test encodes overlapping row strides and differs from the implemented contiguous fixed-point model. Approval of the documented adapted reference is required if literal provided-testbench compliance is graded. The confirmed LED counter does not by itself prove every switch-to-LED mapping.

## Fresh verification and evidence

The expanded ANN flow has passed both native C simulations, baseline and pipelined synthesis, and all five VHDL co-simulation transactions. HLS latency remains 33,056 / 6,612 cycles. Co-simulation latency is 6,610 / 6,612 / 6,613 cycles (min/average/max); the adapted course vector is included. Whole-system Vivado implementation was rebuilt after export and again met timing with WNS +3.462 ns, WHS +0.023 ns and zero negative slack.

### Corrected optimized benchmark

The fresh Vitis build verifies `-O2 -g3` in the actual compiler command. The initial and ten subsequent ZYBO transactions all returned matching custom vector outputs and overall PASS. Repeated measurements use the same inputs:

| Measurement | Min ticks | Mean ticks | Max ticks |
|---|---|---|---|
| CPU reference | 13,844 | 14,373 | 14,757 |
| AXI loading | 228,396 | 228,451 | 228,567 |
| HW start-to-done | 21,623 | 21,632 | 21,636 |
| HW FULL transaction | 253,315 | 253,386 | 253,559 |

At 325 MHz the means are 44.2 / 702.9 / 66.6 / 779.6 $\mu$s. The optimized ARM is 1.51× faster than hardware computation alone; FULL hardware is 17.63× slower than CPU compute. The earlier 8.61× comparison against `-O0` is superseded, not retained as the current conclusion.

FULL includes packing, writes, readiness, start/done, readback and unpacking; it excludes platform/data setup, verification and printing. Timer overhead and cache effects remain; ten repeated samples are not a broad workload study.

### Evidence locations

`outputs/fix_evidence/board_ex3_20261011/`, `board_ex3_counter_repeat_20261011/`, `board_ex4_20261011/` and `board_ex5_20261011/` contain real UART/XSDB logs and artifact hashes. The rerun folder also retains the user's physical counter observation. Native compiler/OCM evidence is in `Assignment_2/hardware/Ex345_vitis/native_build_evidence.json` and each app's `build/ocm_verification.json`, ELF, map and compiler-command database.

New ANN evidence: `outputs/fix_evidence/ex7_hls_fresh.log`, `ex7_vivado.log`, `ex7_vitis_retry.log`, and `board_ex7_20261011/` with UART/XSDB logs and bitstream/ELF hashes. Compiler evidence is also at `Assignment_2/hardware/Ex7_vitis/neuralNetTest/build/build_configuration.txt`.

## Historical review: pre-fix snapshot

The following sections preserve the findings at commit `c38a216` before the authorized fixes. Their words "current", "missing" and "outstanding" refer to that snapshot. The closure summary on the preceding pages gives the updated state and supersedes those status claims without erasing the audit trail.

**Verdict: not yet ready for an unqualified hand-in.** Exercise 7 has genuine native build and physical-board evidence, but several written requirements remain unmet or unverified. The earlier statement that all code-level deliverables were complete was too broad.

## Scope and method

This review checks the original assignment (25 August 2025, pp. 2–12), the installation guide, supplied project/IP archives, current sources, generated reports, and actual ZYBO UART output. Exercises 1, 2 and 6 provide supporting labs; the mandatory journal covers 3, 4, 5 and 7. Audio playback is a separate side project, not a missing mandatory deliverable.

Findings distinguish defects, missing evidence and ambiguous course notation. Application code was not changed. The 11 October board retry reused the existing native bitstream and ELF; it was not a clean rebuild or another Vivado session.

## Priority findings

1. **Exercise 3 targets the wrong LED peripheral.** The supplied platform uses custom `led_ip_0`; the application assumes an AXI GPIO called `leds`. This prevents using the application unchanged on the intended hardware, even in the older SDK flow.
2. **Exercise 3 memory placement and modern GPIO support are incomplete.** No linker/build artifact proves heap and stack in on-chip RAM (OCM). Its GPIO initialization still uses legacy device IDs under a modern SDT flow.
3. **Exercises 4–5 are not demonstrated on the target.** The software fixture passes, but the prescribed matrix interfaces differ and the printed transpose notation needs resolution. Exercise 5's required measured software/hardware comparison is absent.
4. **Exercise 7 does not pass the supplied testbench.** The custom testbench passes; the untouched course testbench fails against the same core. The discrepancy stems from different weight-row indexing and needs an explicit, accepted explanation, not an undocumented replacement.
5. **Acceptance and reporting remain open.** Physical switch/LED observation is missing. The course explicitly specifies 2017.2, while this work uses 2026.1. Performance claims need compiler and timing-boundary qualifiers.

## What is already supported

The matrix fixture and custom ANN host/HLS/RTL tests pass. Native implementation met timing. Physical ZYBO runs on 10 and 11 October produced ten matching software/hardware outputs. These results do not prove the other requirements.

# 1. Exercises 3 and 4: requirements and gaps

| Requirement | Finding | Evidence / closure condition |
|---|---|---|
| Ex3 UART interpreter, commands 1 and 2 | Source logic exists | Parser, switch read and timed 0–15 counter exist; target integration is blocked by LED-driver mismatch. |
| Ex3 correct LED device | **Mismatch** | `main.c:21-28` uses `XGpio`. Supplied BD has custom `led_ip_0` at `0x43C00000`. |
| Ex3 one-second increments | Unverified physically | Uses 325,000,000 timer ticks. Observe the LED sequence and actual timing on the board. |
| Ex3 heap/stack in OCM | **Missing proof** | Supply the linker script, built ELF/map and section addresses, not just a stated intention. |
| Ex4 global matrices | Present | `aInst`, `bTInst` and `pInst` exist. |
| Ex4 prescribed types and functions | **Different interfaces** | Course image specifies `vectorType` and `VectorArray`; current type names and initializer/display signatures differ. |
| Ex4 values and row-dot-row multiply | Partial / ambiguous | Current fixture passes modulo-256 arithmetic; its input orientation differs from a literal reading of p. 6. |
| Ex4 `mult`, UART result and ticks | Source only | No retained Ex4 native build or physical UART measurement. |

### The LED and GPIO issues are separate

The supplied Ex3 archive's `system_bd.tcl` instantiates `xilinx.com:user:led_ip:1.0`. Course lab 4 uses a custom LED-IP register write, and its RTL drives the LEDs from register 0. Replacing the device-ID macro alone will not turn that device into AXI GPIO.

For switches, the installed 2026.1 `xgpio.h` selects a base-address initialization argument under `SDT`. Current Ex3 code passes `XPAR_SWITCHES_DEVICE_ID`; the shared timer already has a conditional legacy/SDT path, but GPIO does not. A compatible fix must respect the actual platform's generated address map.

### Matrix orientation: resolve before changing expectations

Let $C$ denote the array stored in `bTInst`. The implementation computes $P_{ij} = \sum_k A_{ik}C_{jk} \pmod{256}$, i.e. $P = AC^T$. The current $C$ stores constant rows $1, 2, 3, 4$, yielding rows $(10, 20, 30, 40)$, $(26, 52, 78, 104)$, $(42, 84, 126, 168)$ and $(58, 116, 174, 232)$.

The course p. 6 prints a transpose outside that constant-row matrix. Literal transposition gives four stored rows $(1, 2, 3, 4)$ and results with repeated row values $30, 70, 110, 150$. Page 7's notation is also loose. This is an interpretation issue, not evidence that the row-dot-row arithmetic is wrong. The hand-in must state its chosen stored layout and use consistent matrix labels; its current table does not do so.

## 2. Exercise 5: acceleration is not yet measured

The supplied IP takes two packed 32-bit rows in registers 0 and 1 and returns an eight-bit dot product in register 2. Truncating each product before summing and truncating the final sum is equivalent to the software result modulo 256. The current `multiMatrixHard` follows this register protocol.

The actual supplied IP uses `S00_AXI`; the course example uses `S_AXI`. The code's `S00_AXI` spelling matches the supplied package and is not a defect. However, source-level agreement is not a native platform build, a physical result, or a measured comparison.

**Required closure:** build the matrix-IP hardware and application; capture the full software and hardware matrices, equality result and both tick counts over UART. Convert ticks using 325 MHz and discuss the 650 MHz CPU versus 100 MHz PL clocks and the cost of AXI accesses. The existing journal's measurement plan cannot substitute for these results.

## 3. Exercise 7: all nine numbered items

| Item | Requirement | Current evidence / verdict |
|---|---|---|
| 7.1 | C++ IP and testbench | Implemented; custom self-checking reference passes. |
| 7.2 | Simulation/synthesis with provided testbench | **Unmet literally.** Native reports use the replacement testbench. Untouched supplied testbench fails a direct host check. |
| 7.3 | AXI4-Lite arrays | Implemented. Generated control and array buses are both connected and addressed. |
| 7.4 | Pipeline both inner loops; compare latency and FF/LUT/DSP/BRAM | Supported by baseline/pipelined reports. Both use 2026.1 and a 10 ns target. |
| 7.5 | Lowest-latency C/RTL co-simulation | VHDL PASS, 6,610 cycles for the custom test. One transaction does not measure transaction initiation interval. |
| 7.6 | Export VHDL, version 1.0.1 | Successful export command requests 1.0.1. XML stores version 1.0 plus revision 1; this alone is not a failure. |
| 7.7 | Vivado IP integration and pin constraints | Saved BD, XDC, native implementation and physical numeric run support completion for this ZYBO. |
| 7.8 | SDK application verification and physical switch/LED test | Numeric check passes in Vitis-built application. SDK substitution is a version deviation. Physical LED observation remains unverified. |
| 7.9 | Software/hardware timing; separate loading and compute; bottleneck | Measured, with limits: ARM build is `-O0`; load-plus-run subtotal excludes packing and output readback. |

**Additional model requirements:** fixed-point arithmetic, Q15 shifts, hidden-layer ReLU and switch-to-LED assignment are present. Dimensions are compile-time macros; the default 196/32/10 and a temporary 8/3/4 host variant pass the custom test. Only the default configuration has synthesis/RTL/board evidence. The SDK application is independently fixed at the default dimensions.

## 4. Neural-network testing: reconcile the references

The untouched provided `Ex7ANN/testbench.cpp` and current accelerator were compiled together with the installed AMD arbitrary-precision headers. That executable exits 1: all ten comparisons fail. Two representative values:

| Output | Current accelerator | Supplied floating reference |
|---|---|---|
| 0 | 331 | 41.003418 |
| 9 | 4805 | 201.729141 |

The original reference starts input-weight rows at `NEURONS_LEN*i` and output-weight rows at `OUTPUT_LEN*i`. The implemented fully connected model instead uses `INPUT_LEN*i` and `NEURONS_LEN*i`. Those are sensible contiguous row strides; reverting them blindly would overlap rows. The supplied reference also uses floating arithmetic, whereas the implemented reference matches the prescribed integer shifts. These are modeling/test changes, not consequences of a newer IDE.

**Recommended resolution, not an implemented fix:** retain the supplied test as provenance, document the row-stride discrepancy, and add a clearly labeled adapted course reference with justified fixed-point expectations. Run that test through C simulation and RTL co-simulation, and seek course acceptance for the deviation. Passing only the new synthetic fixture does not prove the literal supplied-testbench requirement.

## 5. Verified native results

| HLS solution | Cycles | BRAM18K | DSP | FF | LUT |
|---|---|---|---|---|---|
| Baseline | 33,056 | 10 | 2 | 638 | 716 |
| Pipelined inner loops | 6,612 | 10 | 3 | 1,112 | 1,126 |

The approximately 5× latency reduction costs one DSP, 474 FF and 410 LUT, with unchanged BRAM18K usage. This compares two solutions in the *same* 2026.1 toolchain, not Vivado 2017.2 versus 2026.1.

VHDL co-simulation reports 6,610 cycles. Routed system timing reports WNS +3.462 ns, TNS 0, WHS +0.023 ns and THS 0. Routed system usage is 959 LUT, 1,097 registers, six BRAM tiles and three DSPs. These are whole-system implementation figures, not directly interchangeable with HLS core estimates or the BRAM18K unit count.

The fresh 11 October 2026 physical run downloaded the existing bitstream and ELF through the existing hardware server and returned:

$$[-4,\ 3,\ -2,\ 1,\ -4,\ 2,\ -1,\ 0,\ -3,\ 0].$$

Every hardware value matched the software reference; UART ended with `Overall verification: PASS`. The generated VHDL also contains a direct `outLeds <= inSwitch` connection. This is continuous pass-through, but does not itself prove that someone observed the physical LEDs.

# 6. Timing boundaries and reporting corrections

| Measurement | 10 Oct ticks | 11 Oct ticks | 11 Oct time |
|---|---|---|---|
| ARM software compute | 186,751 | 186,747 | 574.6 $\mu$s |
| AXI input/weight writes | 233,860 | 233,849 | 719.5 $\mu$s |
| Accelerator start-to-done | 21,697 | 21,700 | 66.8 $\mu$s |
| Measured load-plus-run subtotal | 255,557 | 255,549 | 786.3 $\mu$s |

Time is ticks divided by 325 MHz. The fresh compute-only ratio is 186747/21700 $\approx$ 8.61. The application compile command uses `-O0 -g3`; this is a comparison with that unoptimized ARM reference, not a best-possible CPU implementation. Two recorded runs are useful repeat observations, not a statistically characterized benchmark.

Input packing occurs before the loading timer. Hardware execution timing starts before the start-register write and includes done polling; output reads and unpacking happen after it stops. The subtotal therefore excludes preparation, readback and setup. It must not be called full end-to-end inference latency. Even this partial subtotal exceeds software compute, so input/weight traffic is already a demonstrated bottleneck for the current one-shot workflow.

### Corrections needed in the existing hand-in

- Replace "code complete" / "ready for supplied Ex3 platform" with the actual peripheral and linker gaps until corrected and built.
- Limit Ex7's all-checks-passed statement to the custom tests; disclose the provided-testbench failure and required physical LED test.
- Correct matrix labels and explicitly document the stored orientation.
- Qualify the 8.61× number as compute-only versus an `-O0` reference; relabel transfer-plus-compute as a subtotal.
- State the 2017.2 requirement deviation. The course does not merely recommend that version, and acceptance of 2026.1 has not been established.

The original hand-in and code were left unchanged by this review. This document is a review addendum, not a claim that those corrections have already been made.

### Minimum acceptance checklist before submission

1. Resolve tool-version and supplied-reference deviations with the course.
2. Fix Ex3 against the actual custom LED IP, use compatible GPIO lookup, and prove OCM heap/stack placement from the built application.
3. Align the matrix interfaces and clarify inputs; capture Ex3–5 native UART results, physical LED/counter observations and measured Ex5 comparison.
4. Run the documented course-derived neural test and capture physical switch/LED observations. Suggested coverage: patterns 0, 1, 2, 4, 8 and 15.
5. Correct the hand-in's result claims and retain logs with build identity.

These are outstanding assignment actions. Completion of this review is not completion of the assignment itself.

# 7. Evidence index and reproduction

Paths below are relative to the workspace `/home/firas/Documents/Codex/2026-10-08/i-h/`. Repository references use prefix `ERTS_assigments/Assignment_2/`.

| Evidence | Location / what it establishes |
|---|---|
| Course specification | `work/assignment2/Assignment_2/ERTS_Assignment2_ANN_ZYBO.pdf`; pp. 4–9 and 10–12 are the exercise requirements. |
| Installation baseline | `/home/firas/Downloads/ERTS_Vivado_Installation_2017_2.pdf`; pp. 1 and 4 specify 2017.2 and SDK. |
| Ex3 source / platform | `exercise3_console/src/main.c`; original `/home/firas/Downloads/Ex3_ZYBO.xpr.zip`, including `system_bd.tcl` and custom LED RTL. |
| Matrix implementation | `common/matrix.h`, `common/matrix.c`, `exercise4_matrix/src/main.c`, `exercise5_matrix_ip/src/main.c`, `tests/test_matrix.c`. |
| Original ANN reference | `work/assignment2/starter/Assignment_2/Ex7ANN/testbench.cpp`; supplied archive extraction, not the replacement in Git. |
| HLS implementation | `exercise7_neural_net/hls/`; source, header, replacement testbench and `run_hls.tcl`. |
| Native HLS reports | Below that HLS directory: `neuralNet_hls/baseline/syn/report/neuralNet_csynth.rpt`, corresponding `pipelined` report, and `neuralNet_hls/pipelined/sim/report/neuralNet_cosim.rpt`. |
| Vivado integration | `hardware/Ex7/Ex7.srcs/sources_1/bd/system/system.bd`; constraints under `hardware/Ex7/Ex7.srcs/constrs_1/imports/Ex7ANN/contraints.xdc`. |
| Timing / area | `hardware/ex7_timing_summary.rpt`, `hardware/ex7_utilization.rpt`. |
| Application boundaries | `exercise7_neural_net/sdk/neuralNetTest.c:104--153`; packing/loading/run/readback; flags in `hardware/Ex7_vitis/neuralNetTest/build/compile_commands.json`. |
| Physical retry | `outputs/native_screenshots/zybo_retry_20261011T085514Z/uart.log` and `xsdb.log`; fresh hardware evidence. Blank screenshot attempt is excluded. |
| Original recorded run | `outputs/native_screenshots/ex7_board_uart.log`; 10 October benchmark. |

The matrix check was rerun as `make -C Assignment_2 test` from the repository root and passed. Lightweight ANN checks use `g++ -std=c++14` with `-I/home/firas/AMD/2026.1/Vitis/include`. Compile the current core once with the replacement testbench and separately with the untouched course testbench; they exit 0 and 1 respectively. The temporary 8/3/4 check modifies only a copied header, not repository files.

Retained host output is in `outputs/review_evidence/`: the matrix fixture, default custom ANN, alternate 8/3/4 ANN and untouched-course ANN logs. `alternate_neuralNet.h` preserves the temporary dimension settings.

Generated bitstreams, BSPs, ELF files and HLS outputs are largely ignored by Git. Their local presence is evidence for this review, not proof that a clean clone reproduces them. The companion migration journal records the rebuild order, provenance and limitations.